

# HEALTHCARE

## Rural Health Access & Teleconsultation Hub



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# Rural Health Access & Teleconsultation Hub

## Bringing Healthcare Closer to Rural Communities

Find Healthcare

Check Services

Request Consultation

Connect with Doctors

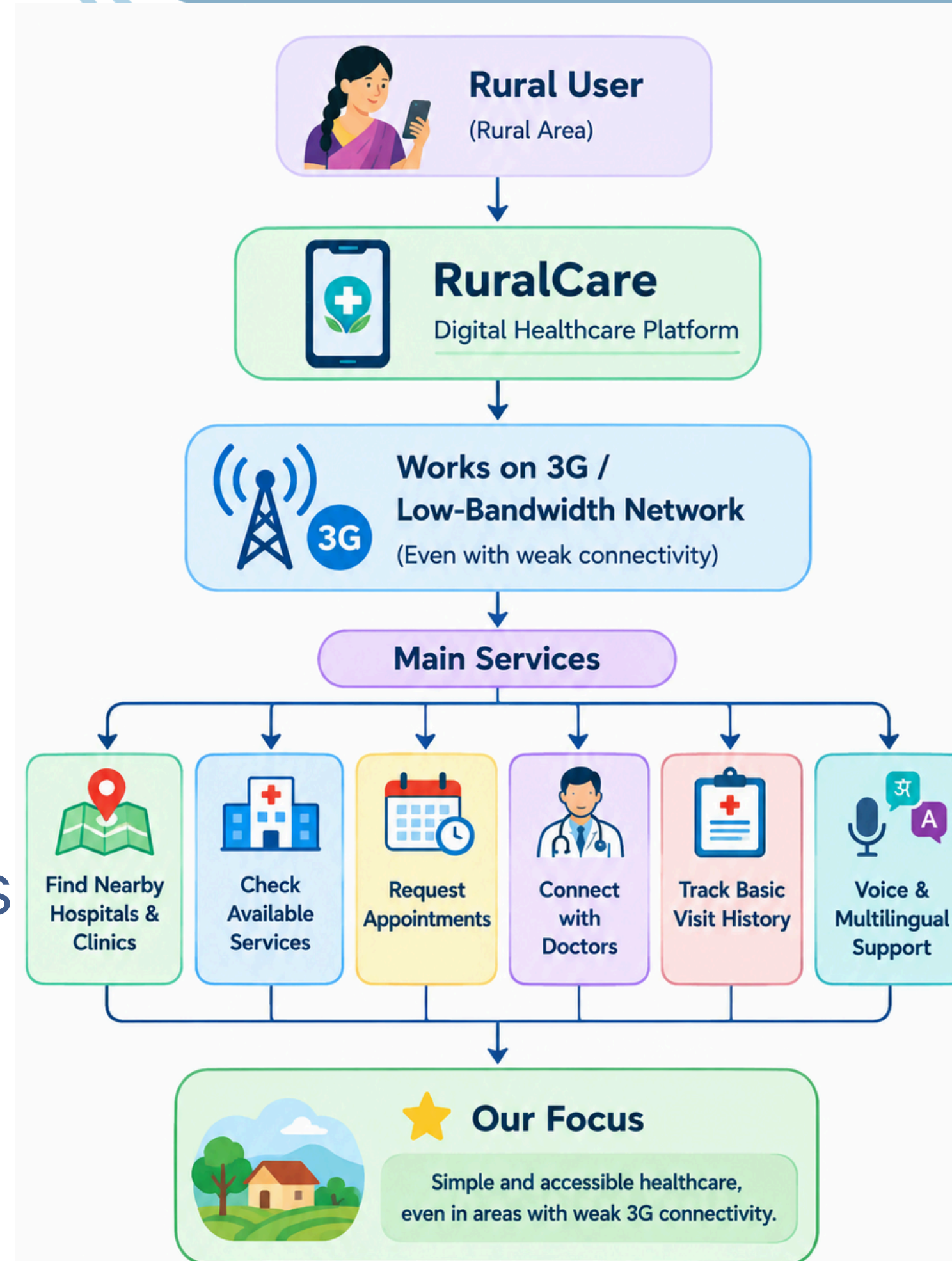
Track Basic Health Records



# OUR SOLUTION

It helps users:

- Find nearby hospitals & clinics
- Check available healthcare services
- Request appointments
- Connect with doctors
- Track basic visit history
- Voice & multilingual support
- Works on 3G & low-bandwidth networks
- Easy healthcare access, even with low internet connectivity.



# EXISTING SOLUTIONS & THEIR GAPS

## How People Access Healthcare Today

- **Direct Hospital Visit**

Problem: Long travel and waiting time.

- **Phone Calls**

Problem: Difficult to know real-time availability.

- **Online Healthcare Platforms**

Problem: Information may be scattered across different platforms.

- **Video Consultation Apps**

Problem: May require stable internet and higher data usage.

- **Online Search**

Problem: Finding the right service can be confusing.



# Our Approach

Designed Around Real Rural Needs

Not just another app,  
but a solution for  
rural healthcare.



## Low Connectivity

Designed for 3G / low-speed networks where available.



## Low Data Usage

Lightweight pages with minimal heavy content.



## Local Access

Find nearby facilities and services.



## Easy Communication

Voice and multilingual support.



## Remote Care

Connect with healthcare providers where connectivity permits.



## Simple Records

Maintain basic patient visit history.

Simple • Accessible • Connected



**Internet detected**



**Connection Check**



**4G / Wi-Fi / Good**

→ Normal website

→ Video consultation



**3G / Moderate**

→ Low Data Mode

→ Compressed images

→ Lightweight pages

→ Low-quality video



**2G / Poor**

**Audio consultation**

**Minimal data**

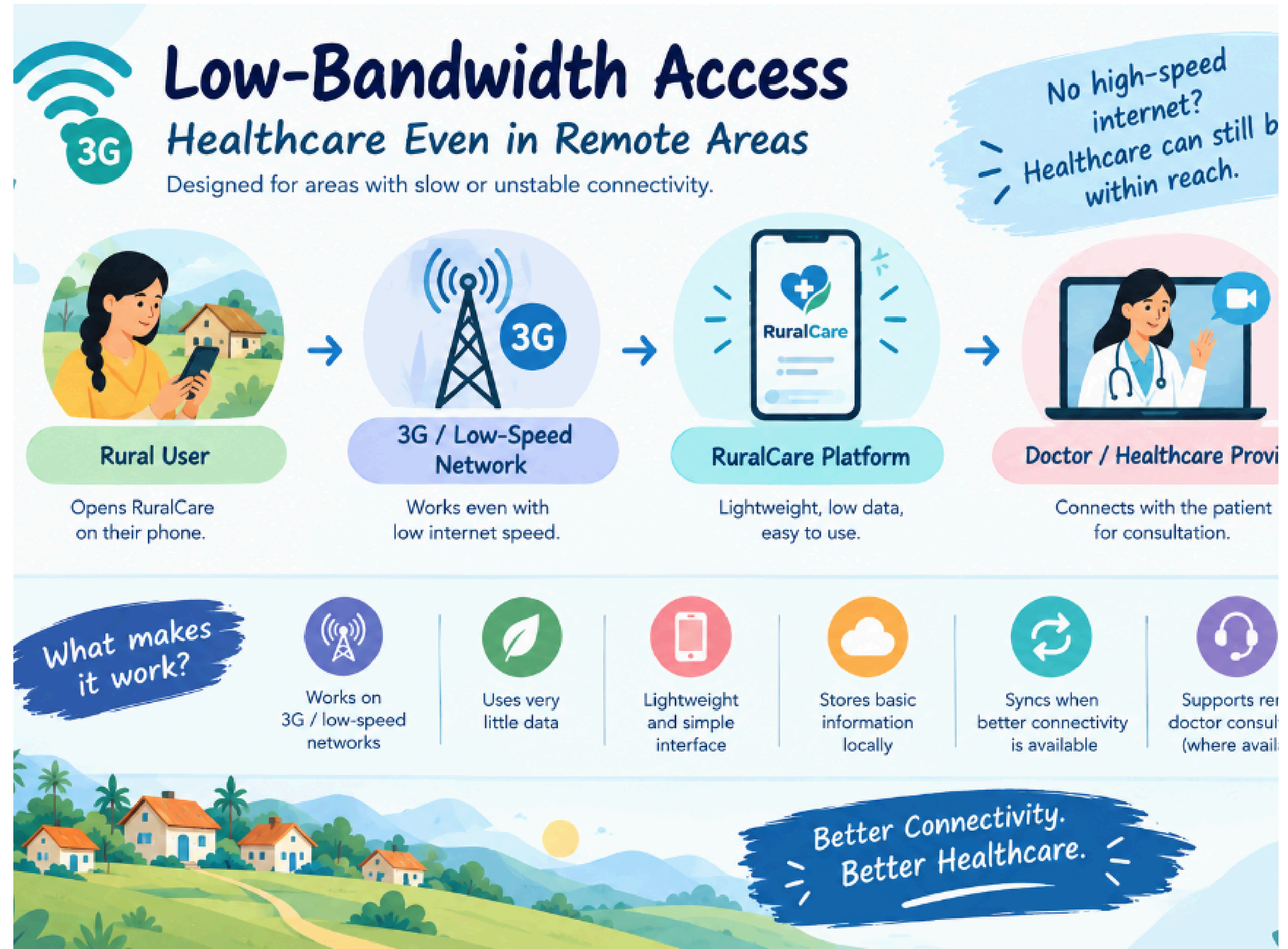


**Offline**

→ Cached pages

→ Save appointment

→ Sync when connection  
returns



# Implementation

**Frontend** – React.js, HTML, CSS

**Backend** – Node.js & Express.js

**Database** – PostgreSQL

**Authentication** – Secure Login & Registration

**Doctor Module** – Doctor profiles and contacts

**Video Consultation** – Real-time doctor–patient calls

**Network Modes** – 4G, 3G, 2G & Offline support

**Deployment** – Host frontend and backend on a live server

# TECHNOLOGY USED

**Frontend:** React.js, HTML, CSS, JavaScript

**Backend:** Node.js, Express.js

**Database:** PostgreSQL

**Authentication:** JWT + bcrypt

**Video Consultation:** WebRTC

**Real-Time Communication:** Socket.IO

**Network Detection:** Browser Network Information API

**API Communication:** REST API

**Deployment:** Cloud hosting / Live Server

# IMPACT & BENEFITS

## Making Healthcare More Accessible

### **For Rural Communities**

- Less unnecessary travel
- Reduced waiting and easily access with 2G & 3G network
- Easier access to healthcare information
- Better communication with providers

### **For Healthcare Facilities**

- Easier appointment management
- Updated service availability
- Better patient communication

### **For Communities**

- Supports underserved areas
- Encourages digital healthcare
- Improves access to available services

# Future scope

- AI Health Assistant – Basic health guidance.
- Smart Network Switching – 4G → 3G → 2G → Offline.
- More Doctors & Hospitals – Expand healthcare network.
- Low-Bandwidth Video Call – Adjusts to network speed.
- Regional Languages – More Indian languages and voice support.
- Reminders – Medicine and appointment alerts.

# Conclusion

RuralCare brings essential healthcare services into one simple platform.

- It helps rural users find nearby facilities, check services, request consultations, and maintain basic records.

Its low-bandwidth design aims to make healthcare access possible even in areas with limited connectivity.

Voice and multilingual support can make the platform easier for different users.

It connects patients and healthcare providers in a more convenient way.



